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LiverTox: Clinical and Research Information on Drug-Induced Liver Injury [Internet]. Bethesda (MD): National Institute of Diabetes and Digestive and Kidney Diseases; 2012-.

Cordyceps

Last Update: June 16, 2025.

OVERVIEW

Introduction

Cordyceps is a genus of fungi (sac fungi) extracts of which have been used in traditional Chinese medicine to treat multiple conditions including fatigue, aging, sexual dysfunction, diabetes, kidney, heart and liver diseases. Cordyceps extracts are generally well tolerated and have not been associated with serum aminotransferase elevations during therapy or with cases of clinically apparent liver injury.

Background

Cordyceps (*sinensis*) is a parasitic fungus that typically infects insect or arthropod larvae that is native to the mountains of Southwest China and Tibet. Cordyceps extracts have been used in Chinese Traditional Medicine for over 1500 years to treat a wide spectrum of medical conditions including fatigue, aging, kidney disease, liver disease, heart disease, diabetes, colitis, and sexual dysfunction. There are more than 260 species of Cordyceps worldwide, the most commonly used in herbal medicine being *Cordyceps sinensis* and *militaris*. Laboratory studies suggest that cordyceps extracts have multiple biologic effects including antioxidant, antiinflammatory, antimicrobial, immunomodulatory, neuroprotective, hypoglycemic, and antineoplastic activities. Despite these findings in laboratory experiments and in animals, none the purported beneficial effects of cordyceps have been shown in clinical trials in humans. *Cordyceps sinensis* extracts contain multiple components including proteins, peptides, amino acids, polyamines, sugars, polysaccharides, mannitol, sterols, organic acids, ergosterol, fiber, vitamins, and minerals. Characteristically, these fungal extracts also contain multiple nucleosides as well as corycepin (3-deoxyadenosine), corycepilic acid, cordyheptaepptide A (a cycloheptaepptide), and bioxanthracenes. The components that account for the biologic activities of cordyceps are not known. Furthermore, most commercial cordyceps supplements in current use are not derived from the natural fungus, but rather from a laboratory, synthetically grown version referred to as *Cordyceps sinensis*-4 [Cs-4]. Because of its purported clinical effects, cordyceps is now found in multiple over-the-counter, commercial forms as capsules or tablets, typically recommended as therapy of fatigue, aging, inflammation, hyperglycemia, and sexual dysfunction, or for boosting energy, athletic performance, immune reactivity, and heart, kidney, liver, or metabolic function. Recommended doses of cordyceps in commercial products range from 0.5 to 4 grams daily. Cordyceps extracts are well tolerated with only minimal or mild adverse events that can include abdominal discomfort, diarrhea, dry mouth, nausea, poor appetite, and rash.

Hepatotoxicity

In multiple short- and long-term clinical trials of different preparations of cordyceps extracts, adverse side effects were described as uncommon and minimal with no mention of either hepatotoxicity or ALT elevations. Few prospective studies have included monitoring of liver tests, but those that did reported no change in serum aminotransferase levels. Despite widespread use, there have been no published reports of serum enzyme elevations or clinically

apparent, acute liver injury attributed to cordyceps extracts. The FDA designates cordyceps as “generally regarded as safe” (GRAS).

A single case report described an elderly woman on long term cordyceps therapy for irritable bowel syndrome who developed hepatportal sclerosis (an uncommon chronic liver condition of unknown cause) which improved once the herbal supplement was discontinued.

Likelihood score: E (unlikely cause of clinically apparent liver injury).

Mechanism of Injury

The mechanism by which cordyceps might cause liver injury is unknown.

Outcome and Management

Acute liver injury from extracts of cordyceps has not been reported.

Drug Class: Herbal and Dietary Supplements

Other common names: Aweto, Caterpillar fungus, Zombie fungus, Cs-4.

PRODUCT INFORMATION

REPRESENTATIVE TRADE NAMES

Cordyceps – Generic

DRUG CLASS

Herbal and Dietary Supplements

CHEMICAL FORMULA AND STRUCTURE

DRUG	CAS REGISTRY NUMBER	MOLECULAR FORMULA	STRUCTURE
Corycepin (Deoxyadenosine)	<u>248010</u>	C ₁₀ H ₁₃ N ₅ O ₃	
Cordyheptapeptide A	<u>11542476</u>	C ₄₉ H ₆₅ N ₇ O ₈	

ANNOTATED BIBLIOGRAPHY

References updated: 16 June 2025

Abbreviations: HDS, herbal and dietary supplements; Cs-4, Cordyceps sinensis-4.

(Expert review of hepatotoxicity published in 1999; several herbal medications are discussed, but not Cordyceps).

Zimmerman HJ. Unconventional drugs. Miscellaneous drugs and diagnostic chemicals. In, Zimmerman, HJ. Hepatotoxicity: the adverse effects of drugs and other chemicals on the liver. 2nd ed. Philadelphia: Lippincott, 1999: pp. 731-4.

(Review of hepatotoxicity of herbal and dietary supplements [HDS] published in 2007; no mention of Cordyceps).

Liu LU, Schiano TD. Hepatotoxicity of herbal medicines, vitamins and natural hepatotoxins.

In, Kaplowitz N, DeLeve LD, eds. Drug-induced liver disease. 2nd ed. New York: Informa Healthcare USA, 2007, pp. 733-54.

<https://en.wikipedia.org/wiki/Cordyceps>

(Wikipedia chapter on cordyceps mentions that it has been widely used in Traditional Chinese Medicine and while in vitro evaluation and animal studies and its chemical constituents suggest potential therapeutic applications, clinical studies have not demonstrated convincing effects in humans).

Parcell AC, Smith JM, Schulthies SS, Myrer JW, Fellingham G. Cordyceps sinensis (CordyMax Cs-4) supplementation does not improve endurance exercise performance. Int J Sport Nutr Exerc Metab. 2004;14:236-42.. [PubMed: 15118196]

(Among 22 male cyclists treated with Cordyceps sinensis [Cs-4, 3 g] or placebo daily for 5 weeks, changes in oxidative capacity and endurance performance were similar in the two groups; no mention of adverse events or changes in serum aminotransferase levels).

Paterson RR. Cordyceps: a traditional Chinese medicine and another fungal therapeutic biofactory? Phytochemistry. 2008;69:1469-95.. [PMC free article: PMC7111646] [PubMed: 18343466]

(Extensive review of Cordyceps species, their classification, preparations, constituents, biologic activities, and purported medical uses; stresses the lack of clinical studies demonstrating the biologic effects in humans and need for future research).

Chen S, Li Z, Krochmal R, Abrazado M, Kim W, Cooper CB. Effect of Cs-4 (Cordyceps sinensis) on exercise performance in healthy older subjects: a double-blind, placebo-controlled trial. J Altern Complement Med. 2010;16:585-90.. [PMC free article: PMC3110835] [PubMed: 20804368]

(Among 15 elderly [ages 50-75 years] adults treated with Cordyceps sinensis [Cs-4: 333 mg] or placebo three times daily for 12 weeks, changes in maximal oxygen uptake and metabolic threshold were similar in the two groups; no mention of adverse events or changes in serum aminotransferase levels).

Jacobsson I, Jönsson AK, Gerdén B, Hägg S. Spontaneously reported adverse reactions in association with complementary and alternative medicine substances in Sweden. Pharmacoepidemiol Drug Saf 2009; 18: 1039-47. [PubMed: 19650152]

(Review of 778 spontaneous reports of adverse reactions to herbals to Swedish Registry found none attributed to cordyceps).

Reuben A, Koch DG, Lee WM; Acute Liver Failure Study Group. Drug-induced acute liver failure: results of a U.S. multicenter, prospective study. Hepatology 2010; 52: 2065-76. [PMC free article: PMC3992250] [PubMed: 20949552]

(Among 1198 patients with acute liver failure enrolled in a US prospective study between 1998 and 2007, 133 [11%] were attributed to drug induced liver injury of which 12 [9%] were due to herbals, including several herbal mixtures, usnic acid, Ma Huang, black cohosh, and Hydroxycut, but not cordyceps).

Teschke R, Wolff A, Frenzel C, Schulze J, Eickhoff A. Herbal hepatotoxicity: a tabular compilation of reported cases. Liver Int 2012; 32: 1543-56. [PubMed: 22928722]

(A systematic compilation of all publications on the hepatotoxicity of specific herbals identified 185 publications on 60 different herbs, herbal drugs and supplements but does not list or mention cordyceps).

Björnsson ES, Bergmann OM, Björnsson HK, Kvaran RB, Olafsson S. Incidence, presentation and outcomes in patients with drug-induced liver injury in the General population of Iceland. Gastroenterology 2013; 144: 1419-25. [PubMed: 23419359]

(In a population based study of drug induced liver injury from Iceland, 96 cases were identified over a 2 year period, 15 of which [16%] were attributed to HDS products, but none were listed as containing cordyceps).

- Bunchorntavakul C, Reddy KR. Review article: herbal and dietary supplement hepatotoxicity. *Aliment Pharmacol Ther* 2013; 37: 3-17. [PubMed: 23121117]
(*Systematic review of literature on HDS associated liver injury does not mention cordyceps*).
- Navarro VJ, Seeff LB. Liver injury induced by herbal complementary and alternative medicine. *Clin Liver Dis* 2013; 17: 715-35. [PubMed: 24099027]
(*Review of the epidemiology, regulatory status, diagnosis, pathogenesis and causes of liver injury from herbal products with specific discussion of conjugated linoleic acid, ephedra, germander, green tea, usnic acid, flavocoxid, aloe vera, chaparral, greater celandine, black cohosh, comfrey, kava, skullcap, valerian, noni juice, pennyroyal and traditional herbal remedies*).
- Navarro VJ, Barnhart H, Bonkovsky HL, Davern T, Fontana RJ, Grant L, Reddy KR, et al. Liver injury from herbals and dietary supplements in the U.S. Drug-Induced Liver Injury Network. *Hepatology* 2014; 60: 1399-408. [PMC free article: PMC4293199] [PubMed: 25043597]
(*Among 839 cases of liver injury from drugs collected in the US between 2004 and 2013, 130 were due to HDS products, including 45 from body building agents [probably anabolic steroids] and 85 from diverse HDS products but no case was attributed specifically to cordyceps*).
- Brown AC. Liver toxicity related to herbs and dietary supplements: Online table of case reports. Part 2 of 5 series. *Food Chem Toxicol* 2017; 107(Pt A): 472-501. [PubMed: 27402097]
(*Description of an online compendium of cases of liver toxicity attributed to HDS products, does not list or discuss cordyceps*).
- Medina-Caliz I, Garcia-Cortes M, Gonzalez-Jimenez A, Cabello MR, Robles-Diaz M, Sanabria-Cabrera J, Sanjuan-Jimenez R, et al.; Spanish DILI Registry. Herbal and dietary supplement-induced liver injuries in the Spanish DILI Registry. *Clin Gastroenterol Hepatol*. 2018;16:1495-1502. [PubMed: 29307848]
(*Among 856 cases of hepatotoxicity enrolled in the Spanish DILI Registry between 1994 and 2016, 32 were attributed to herbal products, the most frequent cause being green tea [n=8] and Herbalife products [n=6], no mention of cordyceps*).
- Jung SJ, Jung ES, Choi EK, Sin HS, Ha KC, Chae SW. Immunomodulatory effects of a mycelium extract of Cordyceps (*Paecilomyces hepiali*; CBG-CS-2): a randomized and double-blind clinical trial. *BMC Complement Altern Med*. 2019;19:77.. [PMC free article: PMC6441223] [PubMed: 30925876]
(*Among 79 adults treated with Cordyceps [1.68 g] or placebo daily for 8 weeks, NK cell activity increased with the herbal treatment but decreased slightly with placebo while serum levels of interleukins, gamma interferon and tumor necrosis factor-alpha did not change and there were no serious adverse events or discontinuations for side effects; no specific mention of serum aminotransferase levels*).
- Liao YH, Chao YC, Sim BY, Lin HM, Chen MT, Chen CY. *Rhodiola/Cordyceps*-based herbal supplement promotes endurance training-improved body composition but not oxidative stress and metabolic biomarkers: a preliminary randomized controlled study. *Nutrients*. 2019;11:2357.. [PMC free article: PMC6835767] [PubMed: 31623349]
(*Among 14 young sedentary adults were enrolled in an exercise training program and treated with a supplement containing Rhodiola crenulate and Cordyceps sinensis or placebo for 8 weeks, changes in maximal oxygen uptake and in serum glucose and insulin were similar in the two groups; no mention of adverse events or serum ALT or AST levels*).
- Kreipke VC, Moffatt RJ, Tanner CJ, Ormsbee MJ. Effects of concurrent training and a multi-ingredient performance supplement containing *Rhodiola rosea* and *Cordyceps sinensis* on

- body composition, performance, and health in active men. *J Diet Suppl.* 2021;18:597-613.. [PubMed: 33078636]
(Among 21 college-aged men treated with an herbal supplement containing cordyceps or placebo for 14 weeks with concurrent aerobic exercise training, body strength improved and there were no overall differences in training performance between the two groups; no mention of adverse events or serum aminotransferase levels).
- Tsai YS, Hsu JH, Lin DP, Chang HH, Chang WJ, Chen YL, Chen CC. Safety assessment of HEA-enriched Cordyceps cicadae mycelium: a randomized clinical trial. *J Am Coll Nutr.* 2021;40:127-132.. [PubMed: 32702252]
(Among 49 healthy adults treated with Cordyceps cicadae extracts [1.05 g] enhanced with hydroxyethyl adenosine [HAE] daily for 3 months, serum ALT, AST, albumin, creatinine, cholesterol and triglyceride levels and complete blood counts did not change).
- Younger J, Donovan EK, Hodgins KS, Ness TJ. A placebo-controlled, pseudo-randomized, crossover trial of botanical agents for Gulf War illness: Reishi mushroom (*Ganoderma lucidum*), Cordyceps (*Urtica dioica*), and Epimedium (*Epimedium sagittatum*). *Int J Environ Res Public Health.* 2021;18:3671. [PMC free article: PMC8037868] [PubMed: 33915962]
(In a crossover study of 30-day courses of high and low doses of three herbal preparations [10 patients each] or placebo, Gulf War illness symptom scales improved with high doses of cordyceps extract daily [1305 mg] compared to placebo [$p=0.048$], but not with low doses [435 mg] or with Reishi mushrooms or Epimedium).
- Ballotin VR, Bigarella LG, Brandão ABM, Balbinot RA, Balbinot SS, Soldera J. Herb-induced liver injury: systematic review and meta-analysis. *World J Clin Cases.* 2021;9:5490-5513. [PMC free article: PMC8281430] [PubMed: 34307603]
(Systematic review of the literature on herb induced liver injury identified 446 references describing 936 cases due to 79 different herbal products, the most common being He Shou Wu [91], green tea [90] Herbalife products [64], kava kava [62] and greater celandine [48]; cordyceps is not listed).
- Bessone F, García-Cortés M, Medina-Caliz I, Hernandez N, Parana R, Mendizabal M, Schinoni MI, et al. Herbal and dietary supplements-induced liver injury in Latin America: experience from the LATINDILI Network. *Clin Gastroenterol Hepatol.* 2022;20:e548-e563. [PubMed: 33434654]
(Among 367 cases of hepatotoxicity enrolled in the Latin American DILI Network between 2011 and 2019, 29 [8%] were attributed to herbal products, the most frequent being green tea [$n=7$], Herbalife products [$n=5$] and garcinia [$n=3$], while cordyceps is not mentioned).
- Kaur B, Vipani A, Trivedi H, Kuo A, Guindi M, Yang JD, Ayoub WS. Traditional Chinese Medicine, Cordyceps, related to hepatportal sclerosis. *ACG Case Rep J.* 2023;10:e01206.. [PMC free article: PMC10686585] [PubMed: 38033616]
(81 year old woman developed abdominal pain and weight loss and was found to have liver disease [bilirubin 0.7 mg/dL, ALT 26 U/L, Alk P 72 U/L, platelet count 76,000/uL] with splenomegaly, small esophageal varices, and a liver biopsy showing hepatportal sclerosis, clinical features improving with stopping use of cordyceps which she had used for 5 years for irritable bowel syndrome).
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(Among 22 healthy long-distance male runners treated with a Cordyceps militaris extract [1800 mg] or placebo daily during 16 weeks of a pre-season training period, blood counts

and serum levels of ALT, AST and creatinine did not change in either group).

Ontawong A, Pengnet S, Thim-Uam A, Munkong N, Narkprasom N, Narkprasom K, Kuntakhut K, et al. A randomized controlled clinical trial examining the effects of Cordyceps militaris beverage on the immune response in healthy adults. Sci Rep. 2024;14:7994.. [PMC free article: PMC10997757] [PubMed: 38580687]

(Among 20 healthy adults given a fermented Cordyceps militaris or placebo beverage once daily for 8 weeks, complete blood counts, serum ALT and AST and immunoglobulin levels did not change in either group; no mention of symptomatic adverse events).

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